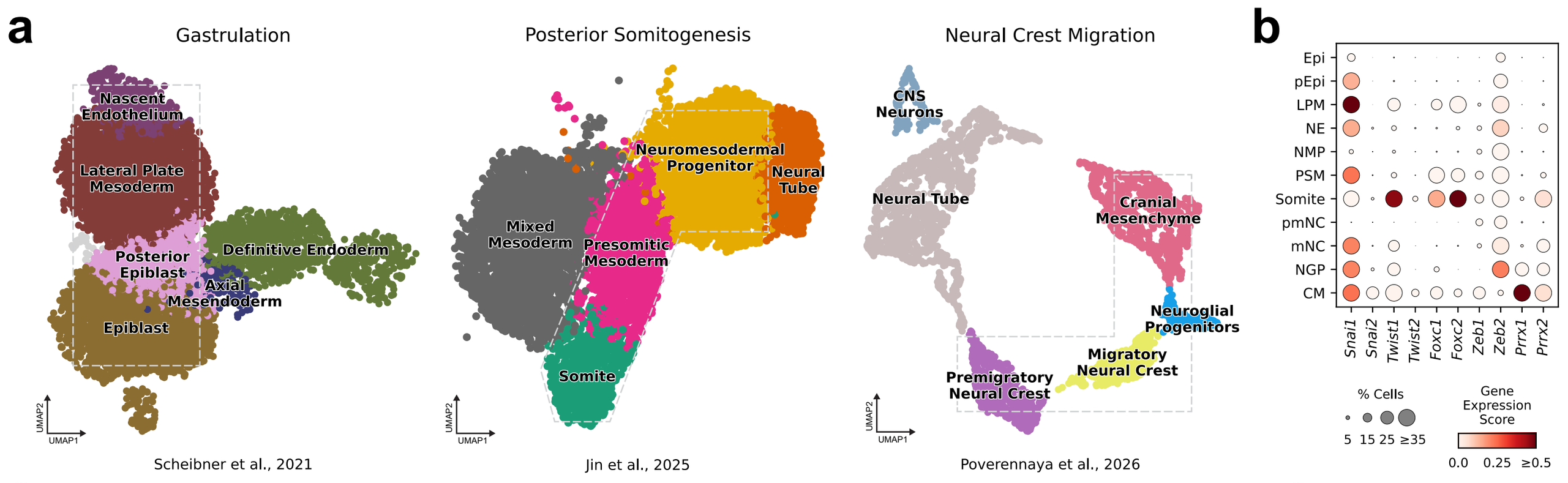

a
Gastrulation
Nascent Endothelium
Lateral Plate Mesoderm
Definitive Endoderm
Posterior Epiblast
Axial Mesendoderm
Epiblast
UMAP2
UMAP1
Scheibner et al., 2021
Posterior Somitogenesis
Neuromesodermal Progenitor
Neural Tube
Mixed Mesoderm
Presomitic Mesoderm
Somite
UMAP2
UMAP1
Jin et al., 2025
Neural Crest Migration
CNS Neurons
Neural Tube
Cranial Mesenchyme
Neuroglial Progenitors
Migratory Neural Crest
Premigratory Neural Crest
UMAP2
UMAP1
Poverennaya et al., 2026
b
Epi
pEpi
LPM
NE
NMP
PSM
Somite
pmNC
mNC
NGP
CM
Snai1
Snai2
Twist1
Twist2
Foxc1
Foxc2
Zeb1
Zeb2
Prrx1
Prrx2
% Cells
5 15 25 ≥35
Gene Expression Score
0.0 0.25 ≥0.5
c
Epiblast
NE
NMP
Somite
pmNC
CM
UMAP2
UMAP1
Max
Min
PseudoTime
e
Gast
pSom
Crest
49
3
21
6
87
8
63
d
Wfdc2
Dnmt3b
Pou3f1
Slc7a3
Utf1
Cldn6
Epcam
Pim2
Fst
T
Mixl1
Sp5
Tdgf1
Frzb
Cyp26a1
Lefty2
Mesp1
Dll3
Lhx1
Fgf3
Fn1
Ccnd2
Ifitm1
Hand1
S100a10
Rbms1
H19
Phlda2
Myl7
Ppic
Crabp2
Slc2a1
Slc2a3
Epha5
Cdx2
Fgf8
Cdx4
Cyp26a1
Ptk7
Hoxaas3
Mest
Rspo3
Msgn1
Cited1
Tbx6
Dll1
Dll3
Fn1
Ifitm1
Meox1
Tcf15
Arg1
Aldh1a2
Foxc2
Fbn2
Ptprt
4933405D12Rik
Fbxw10
1700009N14Rik
Lin7b
Scarna13
Zfp572
Gsdmd
Ccl9
Abcb9
Map3k6
Cnga2
Spata1
1810034E14Rik
Acmsd
Mapkapk3
Max
Min
Scaled Expression
Gast
pSom
Crest
Max
Min
Scaled Signature
f
Top GO Terms
(Biological Process)
Negative Regulation of DNA-templated Transcription (GO:0045892)
Negative Regulation of Transcription by RNA Polymerase II (GO:0000122)
Endodermal Cell Fate Commitment (GO:0001711)
Mesoderm Morphogenesis (GO:0048332)
Regulation of Canonical Wnt Signaling Pathway (GO:0060828)
Generation of Neurons (GO:0048699)
Regulation of Transcription by RNA Polymerase II (GO:0006357)
Canonical Wnt Signaling Pathway (GO:0060070)
Wnt Signaling Pathway (GO:0016055)
Response to Retinoic Acid (GO:0032526)
Telomerase RNA Localization (GO:0090672)
Telomerase RNA Localization to Cajal Body (GO:0090671)
Box C/D snoRNP Assembly (GO:0000492)
RNA Localization to Cajal Body (GO:0090670)
Small Nucleolar Ribonucleoprotein Complex Assembly (GO:0000491)